

Computer Organization & Assembly

Week 8

Memory-1

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Memory

- It is a physical device use for storage purpose.
- Device to store the contents of computer system.
- In computer, we do not use a single memory. We use memory hierarchy.

Memory Hierarchy

- Design constraints on a computer's memory can be summed up by three questions:
 - How much, how fast, how expensive
- There is a trade-off among capacity, access time, and cost
 - Faster access time, greater cost per bit
 - Greater capacity, smaller cost per bit
 - Greater capacity, slower access time
- The way out of the memory problem is not to rely on a single memory component or technology, but to employ a ***“memory hierarchy”***

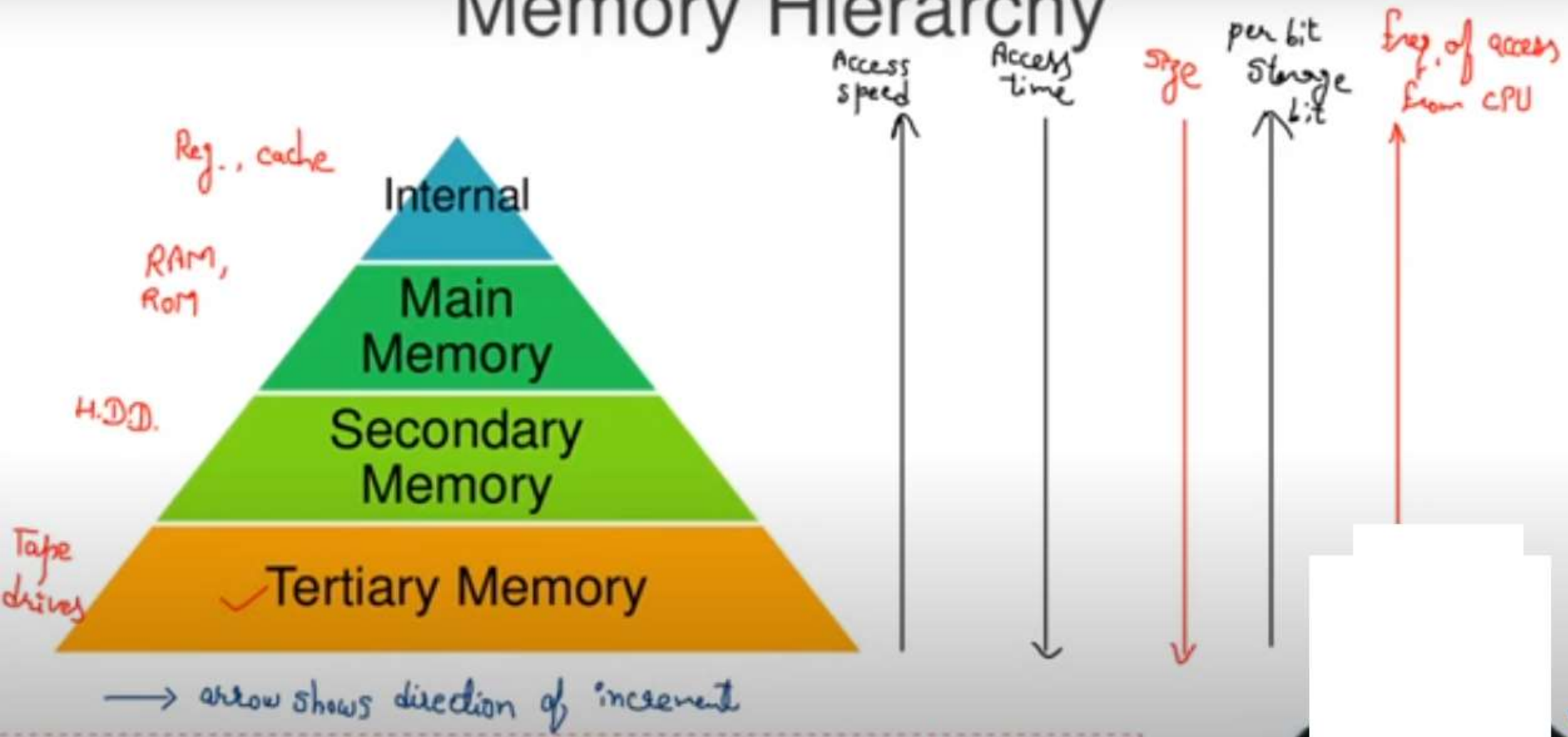
Memory Hierarchy

Memory hierarchy used when discussing performance issues.

Goal of Memory Hierarchy:

- 1. To maximize the Access Speed ✓
- 2. To minimize the Per Bit Storage Cost ✓

Memory Hierarchy



Average Cost of Memory

Memory	Size in Bits	Per Bit storage Cost	Cost of a Memory
M1.	S1	C1	$S1 * C1$
M2	S2	C2	$S2 * C2$
M3	S3	C3	$S3 * C3$

$$S1 + S2 + S3$$

$$\text{Total} = S1C1 + S2C2 + S3C3$$

$$\text{avg per bit mem cost} = \frac{S1C1 + S2C2 + S3C3}{S1 + S2 + S3}$$

$$\text{avg mem cost} = \frac{\sum_{i=1}^k S_i C_i}{\sum_{i=1}^k S_i}$$

S_i = size of memory M_i in bits
 C_i = per bit storage cost of memory M_i .

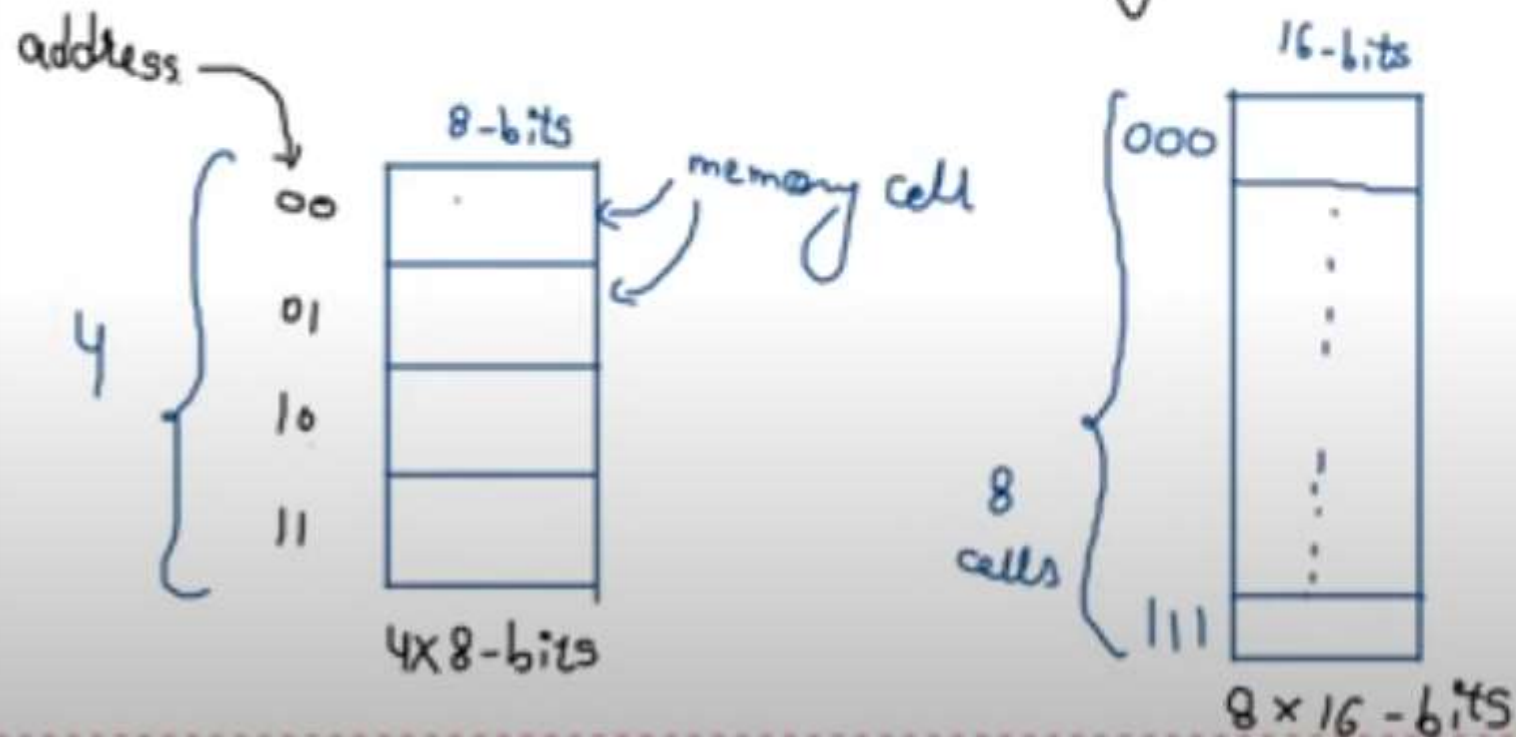
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Memory Representation

Memory is represented by:-

= No. of cells $\times$ 1 cell capacity OR

= No. of memory locations $\times$ bits per location



No. of cells	4	8	16	2^x	n
Add. size	2-bits	3-bits	4-bits	x -bits	$\log_2 n$ bits

Memory Representation

Assume,

128 x 8-bits

or

128 x 1B

or

128B

4GB memory



4GB x 1B

$2^{32} \times 1B$

add. = 32-bits

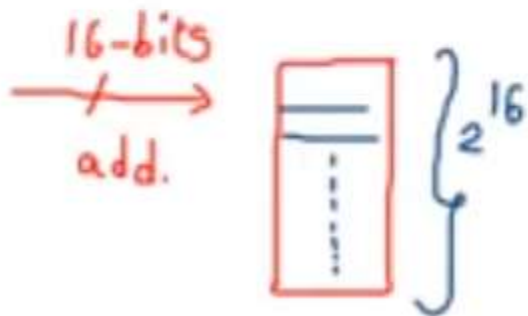
byte addressable.

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Question

A memory has 16-bits address bus. Then how many memory locations are there?

- (A) 64K
- (B) 65536
- (C) 2^{16}
- ☒ (D) All

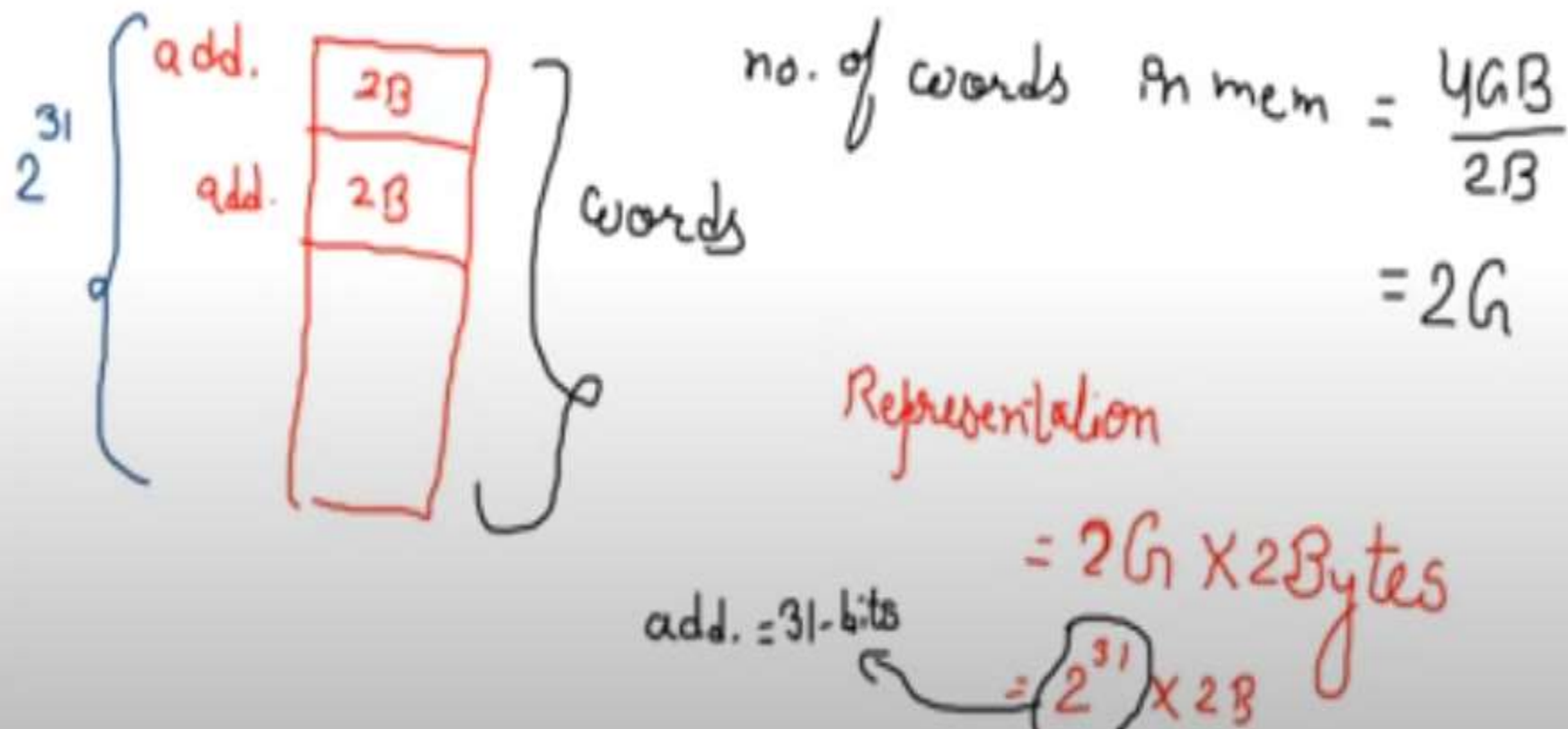


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In the same question, if at each location 16-bits can be stored, then

$$\begin{aligned}\text{Total mem. capacity} &= 2^{16} \times 16\text{-bits} \\ &= 2^{16} \times 2B \\ &= 128KB\end{aligned}$$

A processor can support a maximum memory of 4 GB, where the memory is word-addressable (a word consists of two bytes). The size of the address bus of the processor is at least 31 bits?



Key Characteristics of Computer Memory Systems

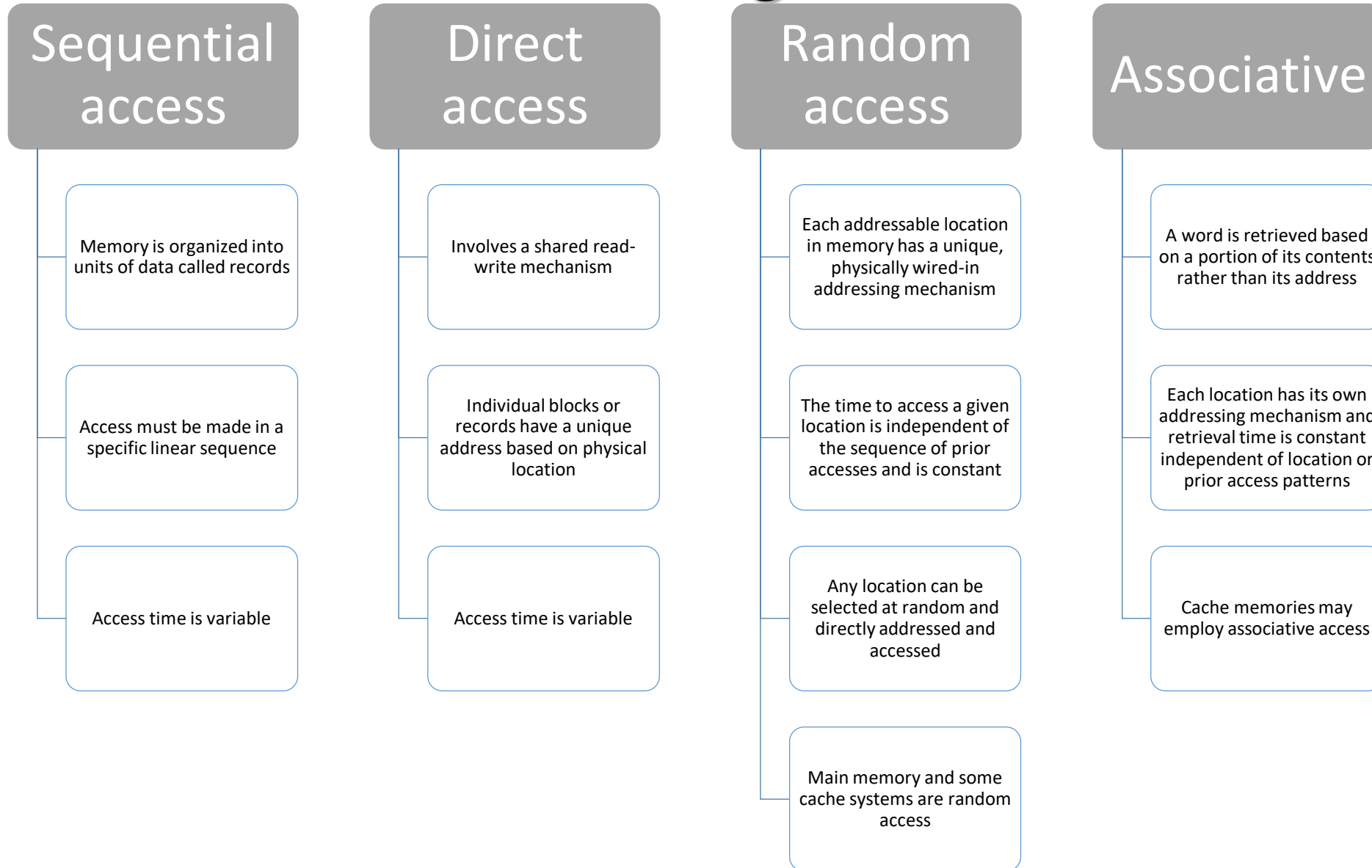
Location Internal (e.g. processor registers, cache, main memory) External (e.g. optical disks, magnetic disks, tapes) Capacity Number of words Number of bytes Unit of Transfer Word Block Access Method Sequential Direct Random Associative	Performance Access time Cycle time Transfer rate Physical Type Semiconductor Magnetic Optical Magneto-optical Physical Characteristics Volatile/nonvolatile Erasable/nonerasable Organization Memory modules
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Table 4.1

Characteristics of Memory Systems

- Location
 - Refers to whether memory is internal and external to the computer
 - Internal memory is often equated with main memory
 - Processor requires its own local memory, in the form of **registers**
 - Cache is another form of internal memory
 - External memory consists of peripheral storage devices that are accessible to the processor via I/O controllers
- Capacity
 - Memory is typically expressed in terms of bytes
- Unit of transfer
 - For internal memory the unit of transfer is equal to the number of electrical lines into and out of the memory module

Method of Accessing Units of Data



Capacity and Performance:

The two most important characteristics of memory

Three performance parameters are used:

Access time (latency)

- For random-access memory it is the time it takes to perform a read or write operation
- For non-random-access memory it is the time it takes to position the read-write mechanism at the desired location

Memory cycle time

- Access time plus any additional time required before second access can commence
- Additional time may be required for transients to die out on signal lines or to regenerate data if they are read destructively
- Concerned with the system bus, not the processor

Transfer rate

- The rate at which data can be transferred into or out of a memory unit
- For random-access memory it is equal to $1/(\text{cycle time})$

Locality Of Reference

- If memory location ***X*** is accessed at time *t*,
- (*assuming no branch instructions*)
- *there is a high probability that*
- memory location ***X + 1*** will also be accessed in the near future.
- This clustering of memory references into groups is an example of

Locality of Reference

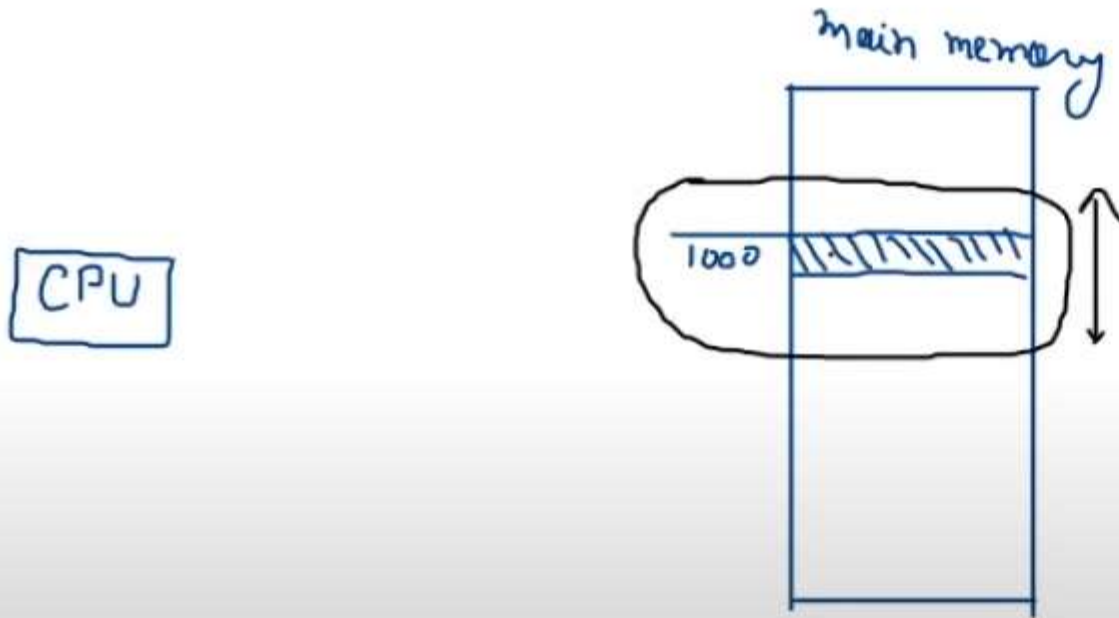
Locality of References

Locality = Neighborhood

Reference = Reference of CPU to Memory (RAM)

To improve the performance of the system we use locality of reference.

If CPU has requested one address for memory access, then that particular address or nearby addresses will be accessed soon.



Locality Of Reference – Types

- ***Temporal locality***

Recently accessed items tend to be accessed again in the near future.

- ***Spatial locality***

*Accesses tend to be clustered in the address space
(for example, as in arrays or loops).*

- ***Sequential locality***

Instructions tend to be accessed sequentially.

Locality Of Reference – Benefit

- *a small amount of very fast memory can be used to effectively accelerate the majority of memory accesses.*

REASON:

- only a small amount of the entire memory space is being accessed at any given time
- values in that space are being accessed repeatedly.
- $\therefore$ values from a slower memory can be copied to a smaller but faster memory that resides higher in the hierarchy.

Locality Of Reference – Benefit

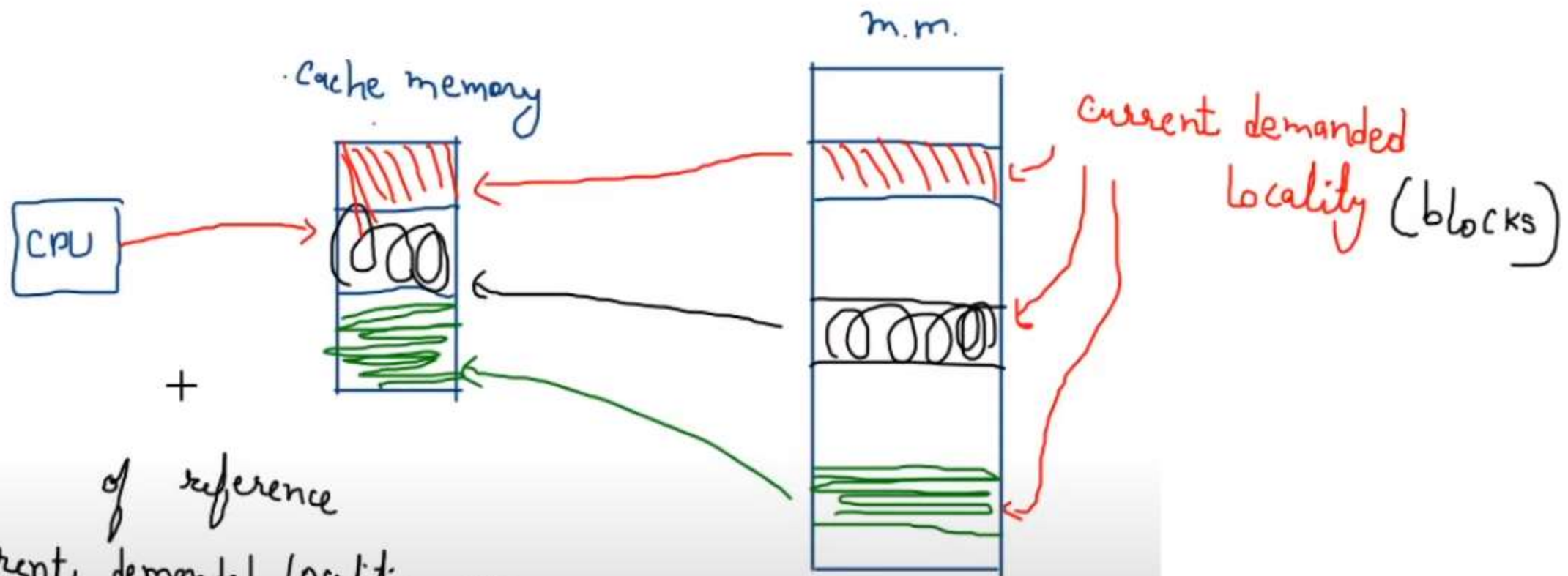
REASON:

- only a small amount of the entire memory space is being accessed at any given time
- values in that space are being accessed repeatedly.
- ∴ values from a slower memory can be copied to a smaller but faster memory that resides higher in the hierarchy.
- **Results in a memory system that can store**
- A large amount of information in a large, low cost memory
- yet provide nearly the same access speeds that would result from using very fast but expensive memory.

Cache Memory

- Cache Memory is
- a relatively small, fast memory
- Interposed between a larger, slower memory and the logic that accesses the larger memory.
- holds recently accessed data
- designed to speed up subsequent access to the same data.
*combines memory access time of
expensive, high-speed memory with
the large memory size of
less expensive, lower-speed memory.*

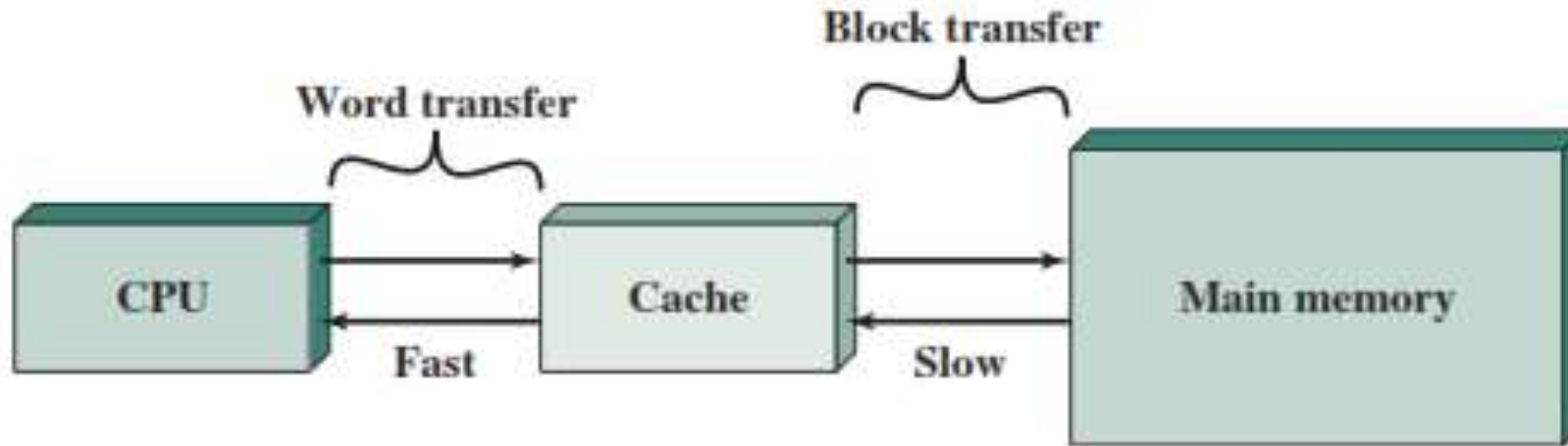
Cache Memory



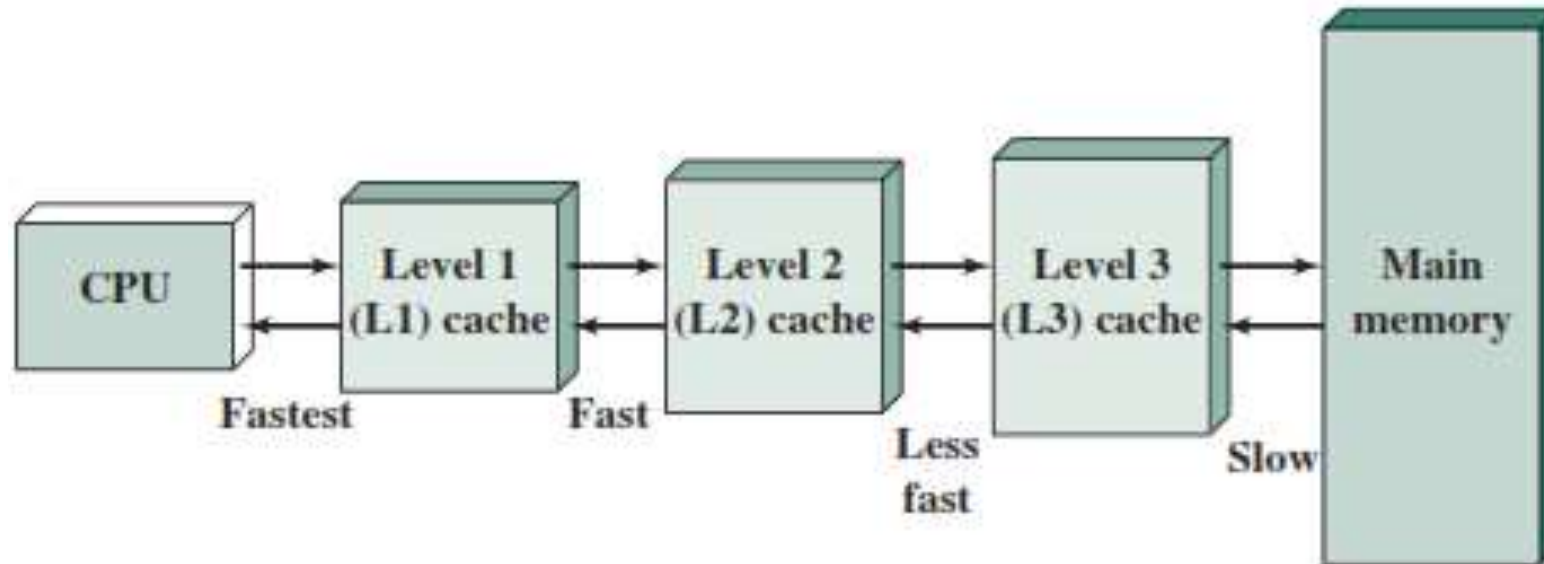
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Based on
concept of reference
current demanded localities are
kept into a smaller & faster memory, known as Cache.

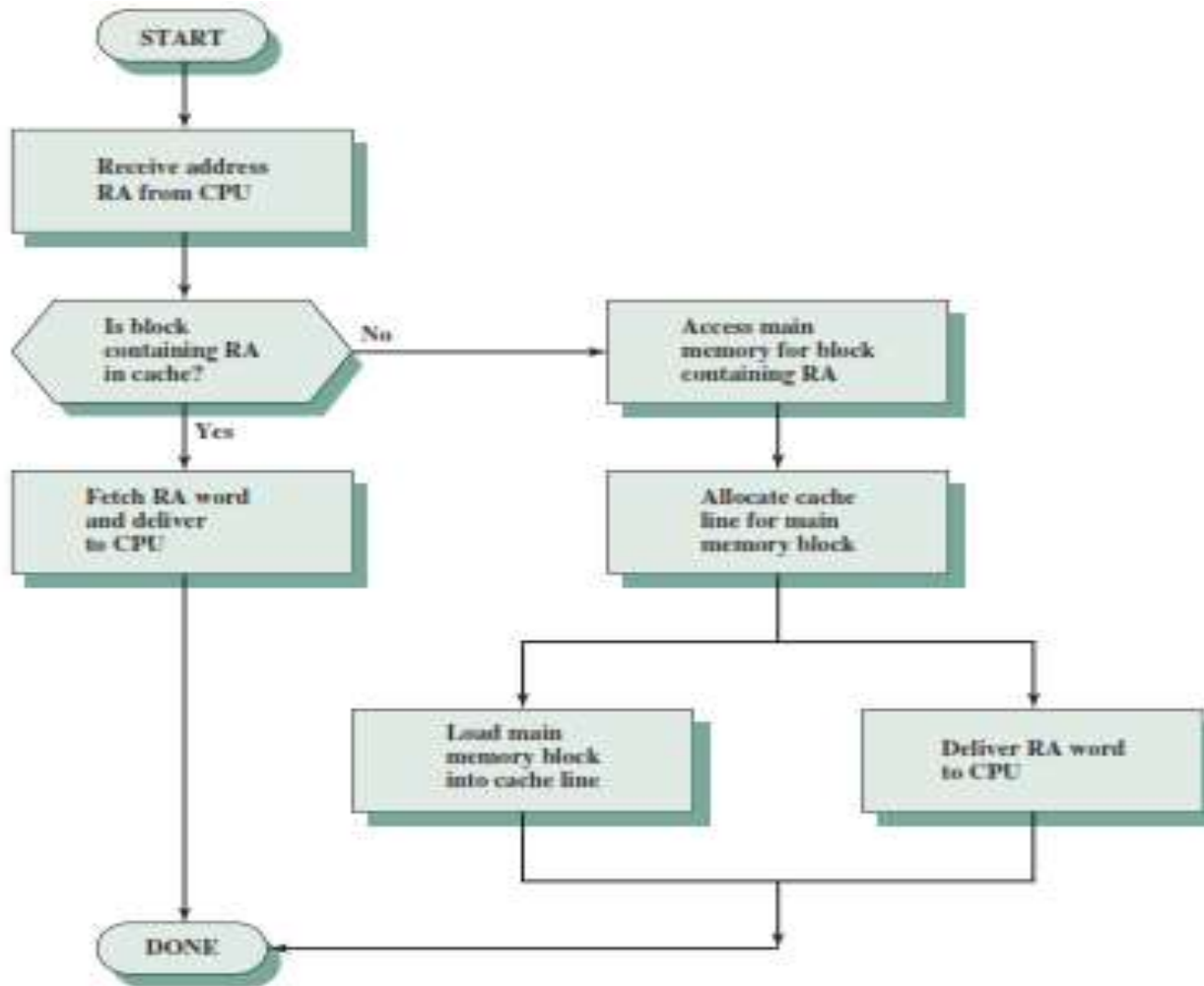
Cache Memory



Cache Memory – Organization



Cache Memory – Read Operation



Elements Of Cache Design

Cache Addresses

- Logical
- Physical

Cache Size

Mapping Function

- Direct
- Associative
- Set associative

Replacement Algorithm

- Least recently used (LRU)
- First in first out (FIFO)
- Least frequently used (LFU)
- Random

Write Policy

- Write through
- Write back

Line Size

Number of Caches

- Single or two level
- Unified or split

Working of Cache Memory

Cache Hit :- If CPU's demanded content is present in CPU.

Cache Miss :- " " " " " is not " " "

Performance of Cache is given by Cache **Hit Ratio**

$$H = \frac{\text{no. of cache hits}}{\text{Total references}} = \frac{\text{hits}}{\# \text{hits} + \# \text{miss}}$$

$$\text{miss ratio} = (1 - H)$$

Average Memory Access Time

⇒ CPU refers memory = 100 times

⇒ No. of hits = 80

⇒ # of misses = 20

Assume,

when hit, mem access time = 10 nsec

" miss, " " " = 100 nsec

Total time for all hits = $80 \times 10 = 800 \text{ ns}$

" " " all miss = $20 \times 100 = 2000 \text{ nsec}$

Total mem. access time = 2800 nsec

for 1 mem access time = $\frac{2800}{100} = 28 \text{ nsec}$
(avg)

+

$$T_{avg} = H * \text{hit time} + (1-H) * \text{miss time}$$

----- (1)

$$= 0.8 * 10 + 0.2 * 100$$

$$= 8 + 20$$

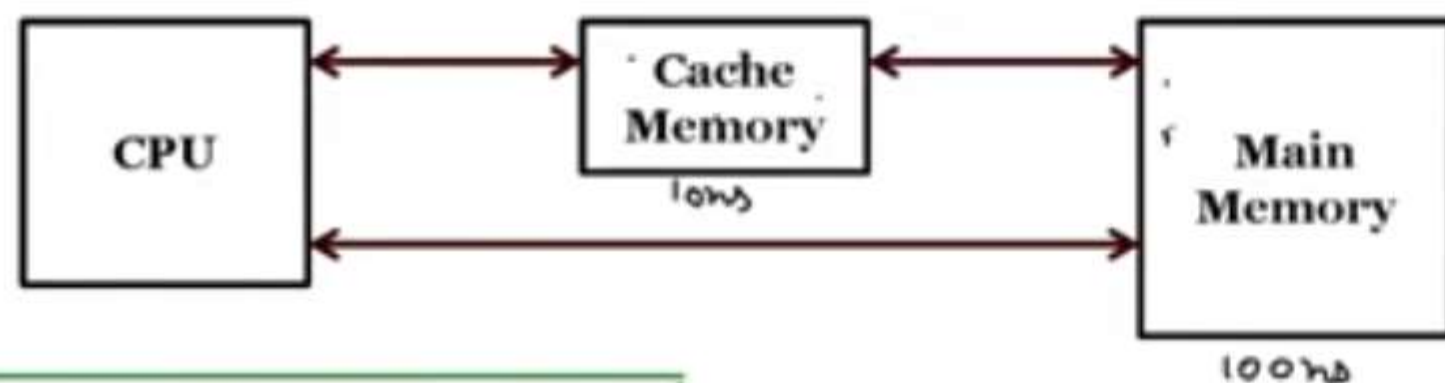
$$= 28 \text{ nsec}$$

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$$\frac{80}{100} * () + \frac{20}{100} * ()$$

Types of Cache Accesses

1. **Simultaneous Access** Request for cache & main mem. are generated simultaneously.



$$T_{avg} = H * t_{cm} + (1-H) * t_{mm} \quad \text{--- (2)}$$

T_{avg} = Avg. mem. access time

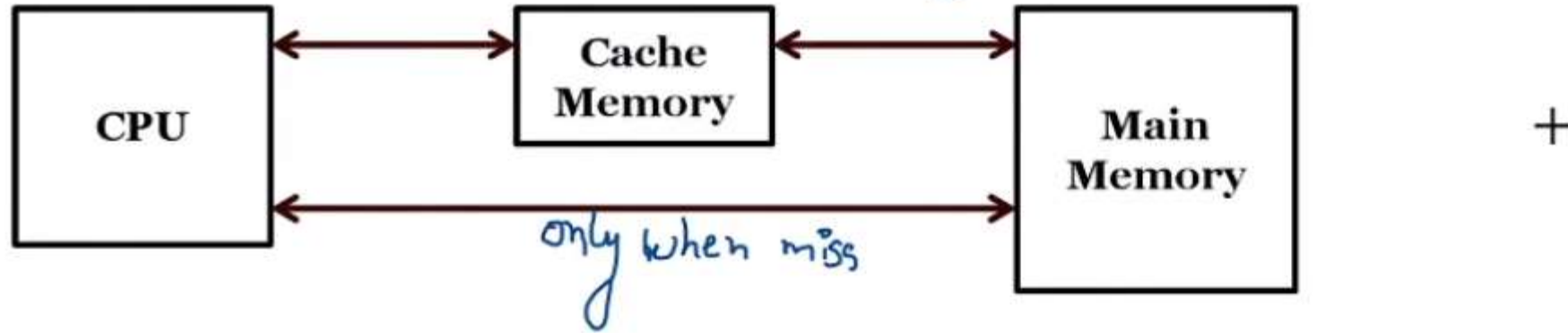
H = Hit ratio

t_{cm} = cache " "

t_{mm} = main " "

Types of Cache Accesses

2. **Hierarchical Access** :- only faster memory is accessed first

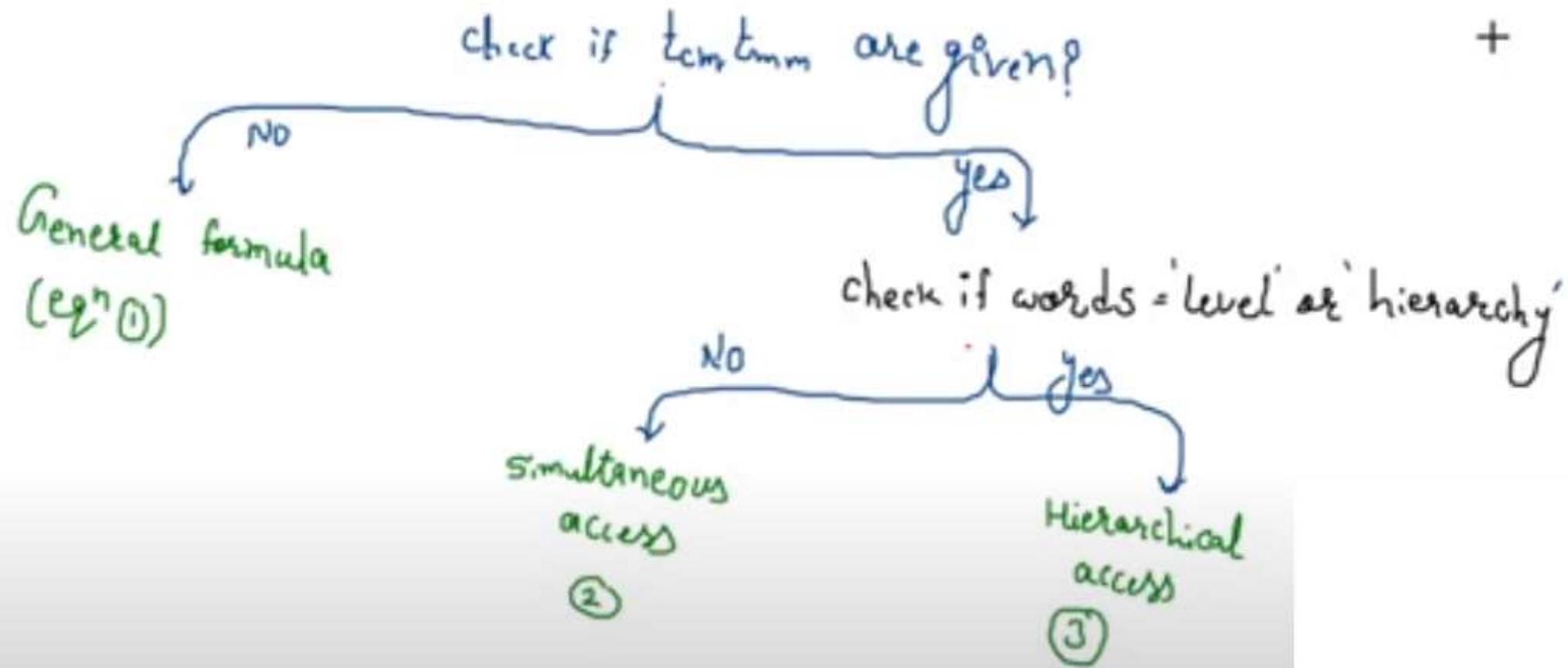


$$T_{avg} = H * t_{cm} + (1-H) (t_{cm} + t_{mm})$$

$$= H t_{cm} + t_{cm} - H t_{cm} + (1-H) t_{mm}$$

$$T_{avg} = t_{cm} + (1-H) t_{mm} \dots \dots \textcircled{3}$$

which formula to use & when?



Example

Assume that for a certain processor, a read request takes 50 nanoseconds on a cache miss and 5 nanoseconds on a cache hit. Suppose while running a program, it was observed that 80% of the processor's read requests result in a cache hit. The average read access time in nanoseconds is _____

General formula

$$\begin{aligned} t_{avg} &= 0.8 * 5 + 0.2 * 50 \\ &= 4 + 10 \\ &= 14 \text{ nsec} \end{aligned}$$

+

Example 2

In a two-level hierarchy, if the ^{nearest to CPU} top level has an access time of 10 ns ^{t_{cm}} and the bottom level has an access time of 60 ns ^{t_{bm}} , what is the hit rate on the top level required to give an average access time of 15 ns ^{t_{avg}} ?

Hierarchical access:-

$$15 = 10 + (1-H)60$$

$$5 = 60 - 60H$$

$$H = \frac{55}{60}$$

$$= 11/12 = 0.91\bar{6}$$

In previous question if hit rate of the top level memory is 100%, then the access time of bottom level memory will be 60 ns?

$$t_{avg} = t_{cm}$$

+

T_{avg} When Locality of Reference Included

Simultaneous:-

$$T_{avg} = H * t_{cm} + (1-H) * t_{block}$$

t_{block} = block access time
from mm.

$$= \text{block size} * t_{mm}$$

Hierarchical:-

$$T_{avg} = t_{cm} + (1-H) (t_{block})$$

+

T_{avg} When Block Transfer Included

Simultaneous:-

$$T_{avg} = H * t_{cm} + (1-H) (t_{block} + \underline{t_{cm}})$$

→ To copy the block into cm.

Hierarchical:-
+

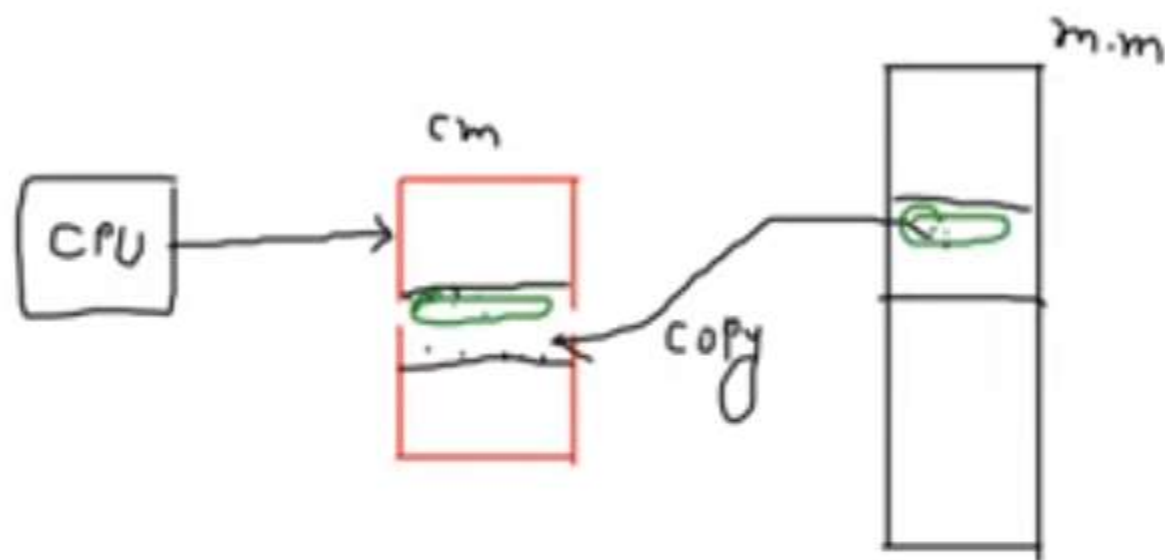
$$T_{avg} = t_{cm} + (1-H) (t_{block} + t_{cm})$$

or

$$= H * t_{cm} + (1-H) (t_{cm} + t_{block} + t_{cm})$$

Note:- In 1 cache access 1 block can be accessed.

Cache Write or write Propagation



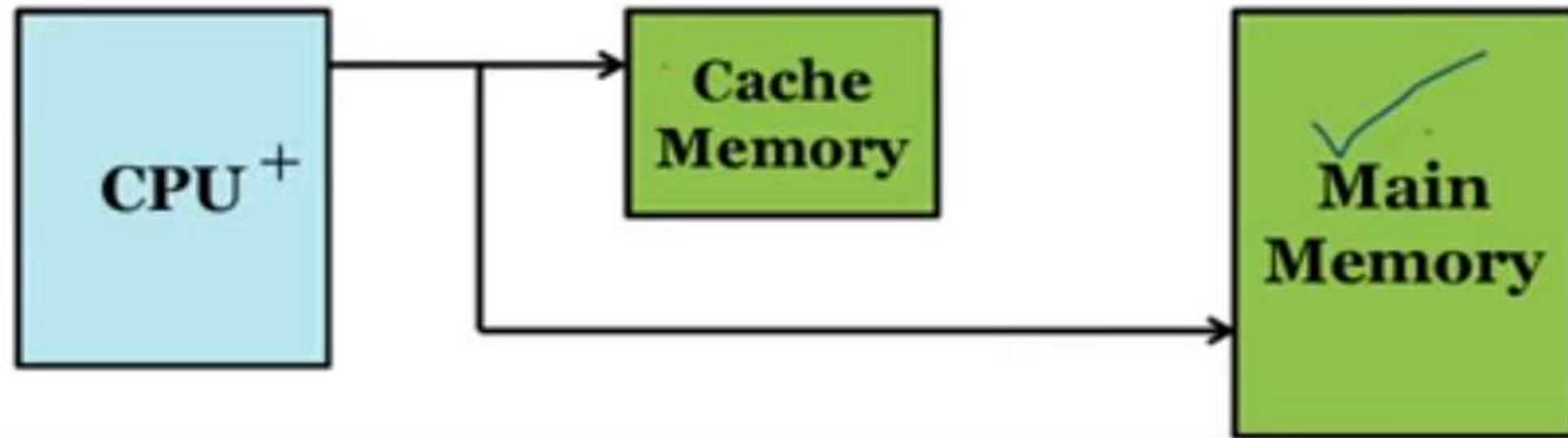
→ write Through

→ write Back

+

if CPU update cm content then the original content in m.m. should also be updated.

Write Through



as soon as CPU updates cache mem content, simultaneously m.m. content is also updated.

T_{avg} in Write Through

$$T_{avg \text{ read}} = H * t_{cm} + (1-H) t_{mm} \quad (\text{simultaneous})$$

$$= t_{cm} + (1-H) t_{mm} \quad (\text{hierarchical})$$

$$T_{avg \text{ write}} = \max(t_{cm}, t_{mm}) = t_{mm} \quad +$$

$$T_{avg} = \text{fraction of read} * t_{avg \text{ read}} + \text{fraction of write} * t_{avg \text{ write}}$$

Example-3

A system has a write through cache with access time of 100ns and hit ratio of 90%. The main memory access time is 1000ns. The 70% of memory references are for read operations.

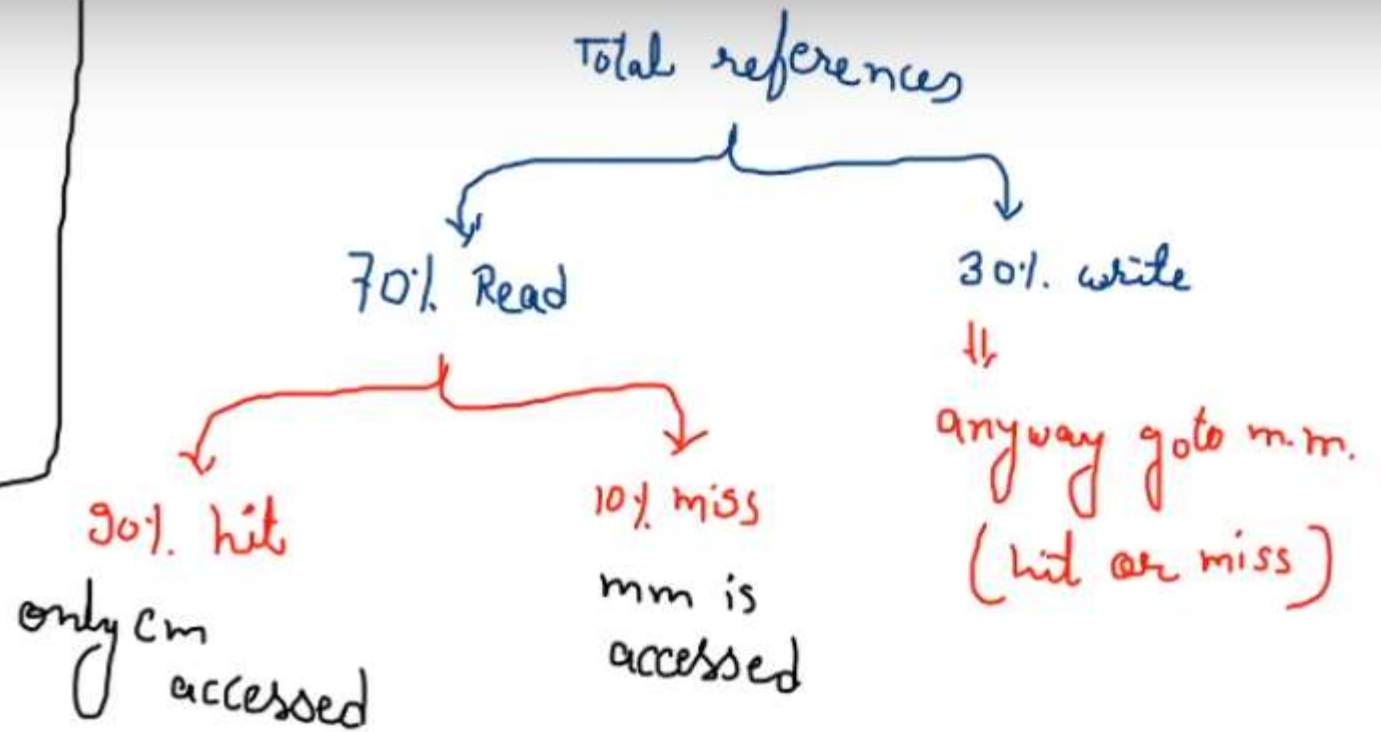
1. Average memory access time for read operations only
2. Average memory access time for write operations only
3. Average memory access time for read-write operations both
4. Effective Hit ratio

$$\begin{aligned} & 0.63 * 100 + 0.37 * 1000 \\ &= 63 + 370 \\ &= 433 \text{ nsec.} \end{aligned}$$

$$\checkmark T_{avg \text{ read}} = 0.9 * 100 + 0.1 * 1000 = 190 \text{ nsec}$$

$$T_{avg \text{ write}} = \max(100, 1000) = 1000 \text{ nsec}$$

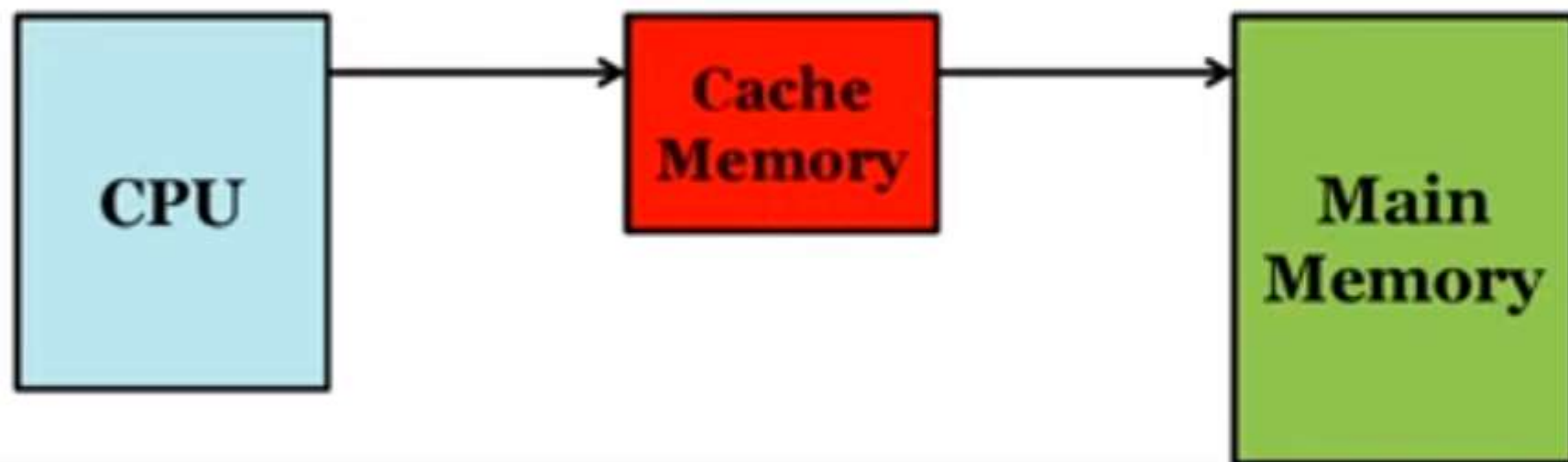
$$\begin{aligned}
 T_{avg} &= 0.7 * 190 + 0.3 * 1000 \\
 &= 133 + 300 \\
 &= 433 \text{ nsec.}
 \end{aligned}$$



Effective hit ratio = % of time CPU's request fulfilled only from cache.

$$= 0.7 * 0.9 = 0.63 \text{ or } 63\%$$

Write Back⁺



CPU only updates in cache & when a block is replaced from c.m. then the block is copied back to m.m.

T_{avg} in Write Back

Simultaneous:-

$$t_{avg \text{ read or } t_{avg \text{ write}}} = H * t_{cm} + (1-H) (t_{block} + t_{write-back})$$

Hierarchical:-

$$t_{avg \text{ read or } t_{avg \text{ write}}} = H * t_{cm} + (1-H) (t_{cm} + t_{block} + t_{write-back})$$

$$t_{write-back} = x * t_{block} \quad x = \text{Fraction of dirty block}$$

Example-4

The access times of the main memory and the Cache memory, in a computer system, are 500 nsec and 50 nsec , respectively. It is estimated that 80% of the main memory request are for read the rest for write. The hit ratio for the read access only is 0.9 and a write-through policy (where both main and cache memories are updated simultaneously) is used. Determine the average time of the main memory (in ns)

$$t_{mm} = 500 \text{ nsec}$$

$$t_{cm} = 50 \text{ nsec}$$

$$\% \text{ of read} = 80\%$$

$$\% \text{ of write} = 20\%$$

$$H_{\text{read}} = 0.9$$

$$\begin{aligned} t_{\text{avg read}} &= 0.9 * 50 + 0.1 * 500 \text{ nsec} \\ &= 45 + 50 \\ &= 95 \text{ nsec} \end{aligned}$$

$$t_{\text{avg write}} = \max(50, 500) = 500 \text{ nsec}$$

$$\begin{aligned} t_{\text{avg}} &= 0.8 * 95 + 0.2 * 500 \\ &= 76 + 100 = 176 \text{ nsec} \end{aligned}$$